Automated Text Classification

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1. Abstract

An investigation examines the discussion topics about Natural Language Processing (NLP) found in Stack Overflow through data collection and analysis. The Stack Exchange API allowed researchers to obtain 20,000 posts which used the [nlp] tag. Each post contained its title with description along with its tags, accepted concluding answer, Question created and answer created date columns. The data received initial processing to make words lowercase and dropped stop words, html tags, punctuation and created tokens and removed all special characters. The analysis included statistical data points showing unanswered posts together with average response duration times which were presented through visual elements. The Word Cloud analyzed NLP terminology after removing nonrelated words.

2. Introduction

The field of artificial intelligence Natural Language Processing (NLP) has now efficiently supports and develops machine translation alongside pdf text summarization as well as sentiment analysis software and chatbot operations. Development teams require Stack Overflow and its analogous communities as crucial sources to obtain NLP solutions because they rely on these platforms as their main solution retrieval platforms. A study examines NLP community methods and real-world NLP problems using posts tagged [nlp] on Stack Overflow. The Stack Exchange API provided 20,000 posts and Data Explorer acquired an incomplete number below 3,000 posts because of implementation restrictions. The data contains several groups including post identifiers, their title content and body, tags, accepted solution content, timestamps and information about individual posts. Our text preprocessing stage first lowered all text to lowercase while also removing particular vocabulary words before the process split words and removed specialized characters with HTML elements. Statistical methods and WordCloud designs helped uncover main NLP concepts in the analysis. A combination of the NLTK model and unsupervised system was used for manual tagging to identify NLP community member priorities and subject matters.

3. Data Collection

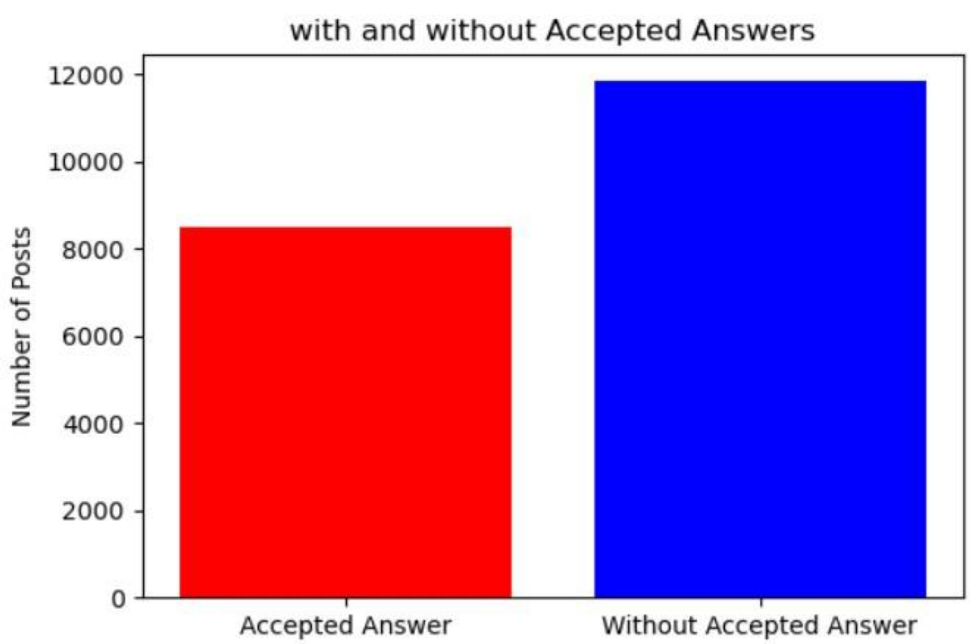
The study data collection focused on obtaining a detailed set of Stack Overflow NLP-related posts. My first plan involved using the Stack Exchange API to retrieve the posts that incorporated the [nlp] tag. The method produced less than 3000 posts while the necessary dataset demanded 20000 posts. After multiple attempts of using API, I was restricted from sending an
API request for 23 hours. The Stack Exchange API struggled with query examination limitations and page break boundaries which impeded my ability to obtain a bigger collection of suitable posts no matter how I changed parameters for page layout and search variables.

I used Piazza for issue discussion before attempting various alternative methods such as different API approaches as well as web scraping techniques. The alternative data retrieval methods also presented challenges because they experienced restrictions in accessing complete datasets.

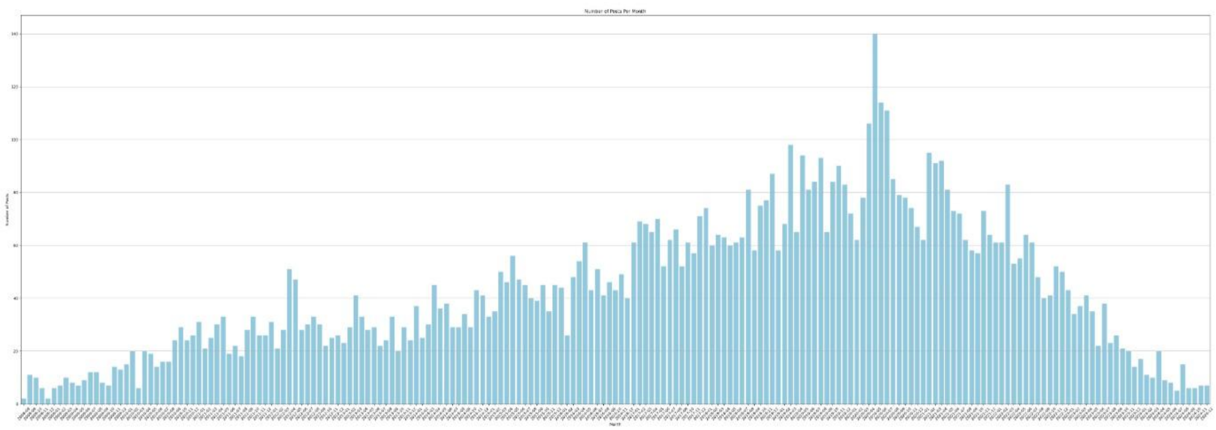
The Stack Exchange Data Explorer (SEDE) became my ultimate solution because it provides free SQL querying functionality for Stack Overflow public data dumps. A purpose-built SQL query allowed me to draw 20,000 posts from Stack Exchange including their [nlp] tags and the corresponding titles, descriptions, tags, and accepted answers. The collection method delivered effective data retrieval for project requirements in a reliable and efficient way. The processed data received its final format as a CSV file which moved to preprocessing steps followed by analysis.

4. Exploratory Data Analysis

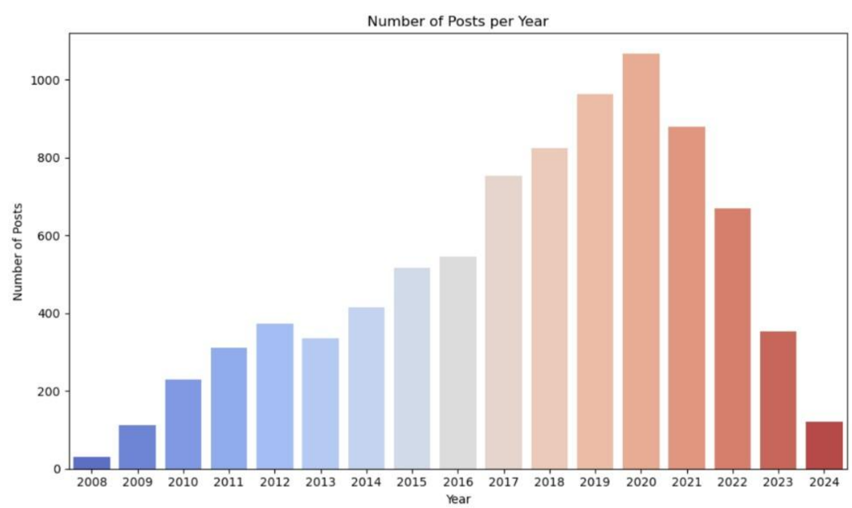
Several visual tools were developed during exploratory data analysis to advance comprehension about the framework and conduct of the database. The initial visualization displayed a bar chart which showed the question counts between those with accepted answers and those without. A basic yet useful chart indicated the data distribution between solved questions and those without resolution in the dataset. The bar chart presented essential data about community answer participation and showcased the number of unresolved questions in the dataset.

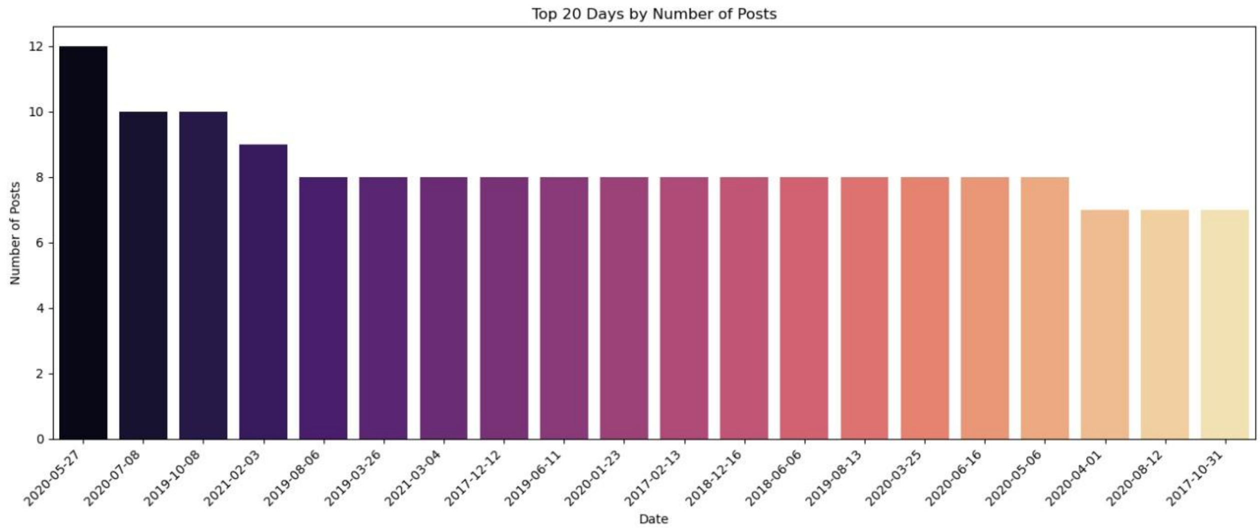


A crucial graphical presentation demonstrated the monthly quantity of posts that user created. The analysis demonstrated posting trends through time-based visualization of Question Created DateTime values separated by month combined in a bar chart. This data representation enabled researchers to detect times when the community activity surged and months when posting frequency reduced. These patterns often reveal actual world events regarding user engagement patterns as well as community growth or outside events that affect platform participation levels.

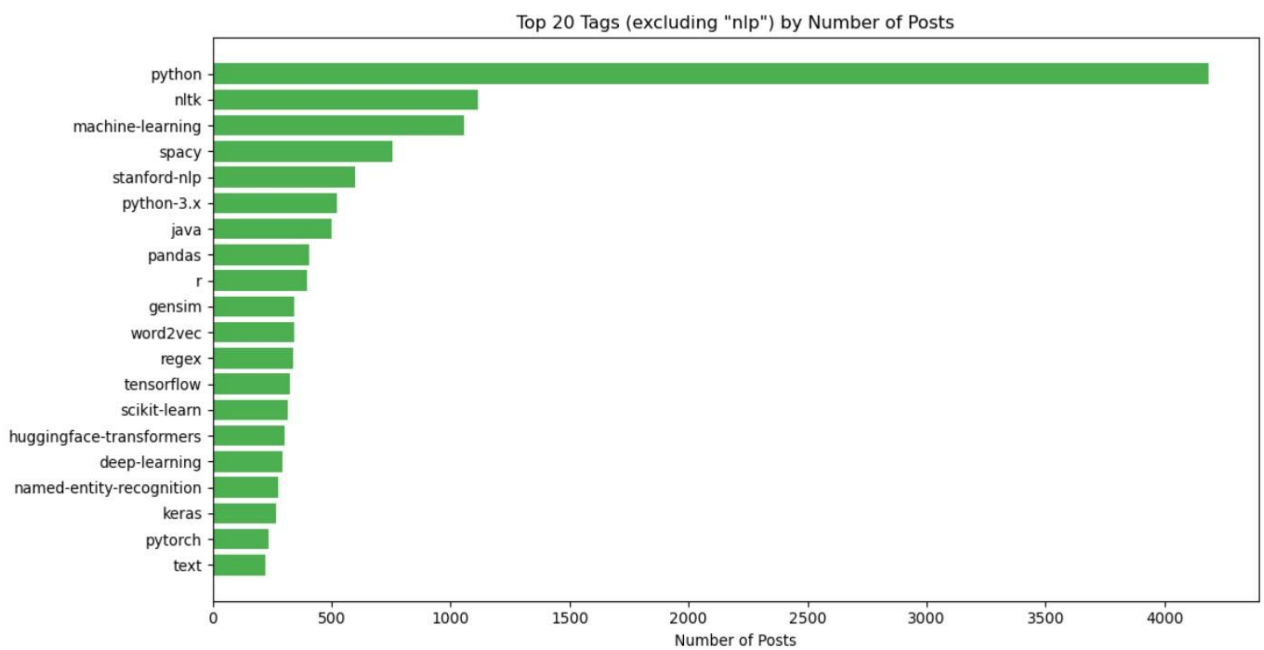


Different visual representations existed alongside each other to track post activity at daily and yearly timescales. The wider trends across months were observed through monthly analysis but daily patterns enabled the detection of rapid activity changes. Yearly post counts offered platform developers an extensive overview of their platform's ups and downs through periods of time across different years. The visual representations enabled researchers to conduct time-based examinations of platform usage at different levels of magnitude.

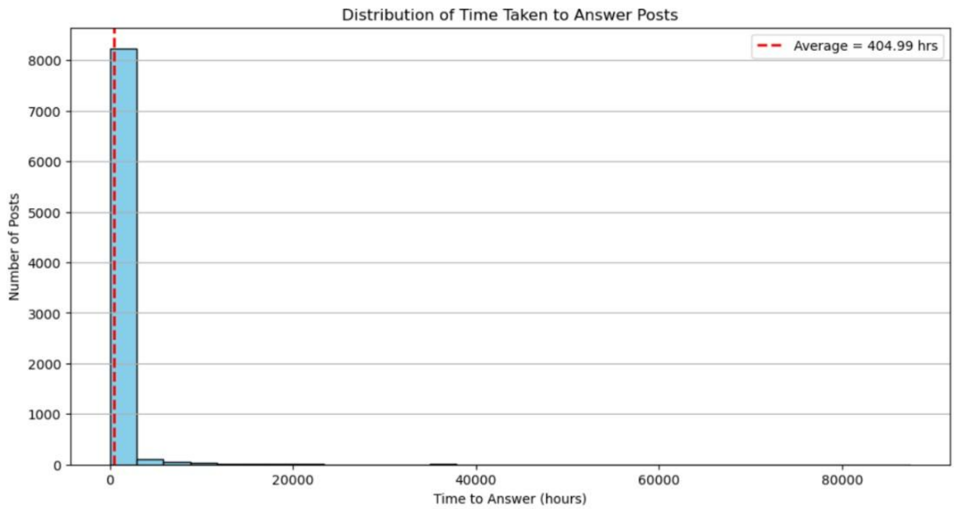




Post tag frequencies were analyzed, and twenty most used tags were identified through frequency analysis. The study highlighted what subjects and technical fields members primarily discussed among themselves. The analysis displayed these frequently used top tags to understand user interests that led to decisions about both topic-centered analysis and platform feature enhancement.



The analysis of dataset responsiveness carried out through measuring the duration between question posting and response receipt showed the average response time. The numeric display of this indicator provided a significant quantitative measure to track community responsiveness despite showing its values without charts. The duration from when a question gets asked to when a user receives a response shows active user engagement but may suggest expertise delays or voids on specific topic areas.



There was a combination of visual representations which described structural patterns of the dataset as well as revealing significant behavioral patterns and usage trends across users and content popularity levels. The team created necessary baseline knowledge about platform functions and social community dynamics through time so decision-making could benefit from it.

5. Preprocessing

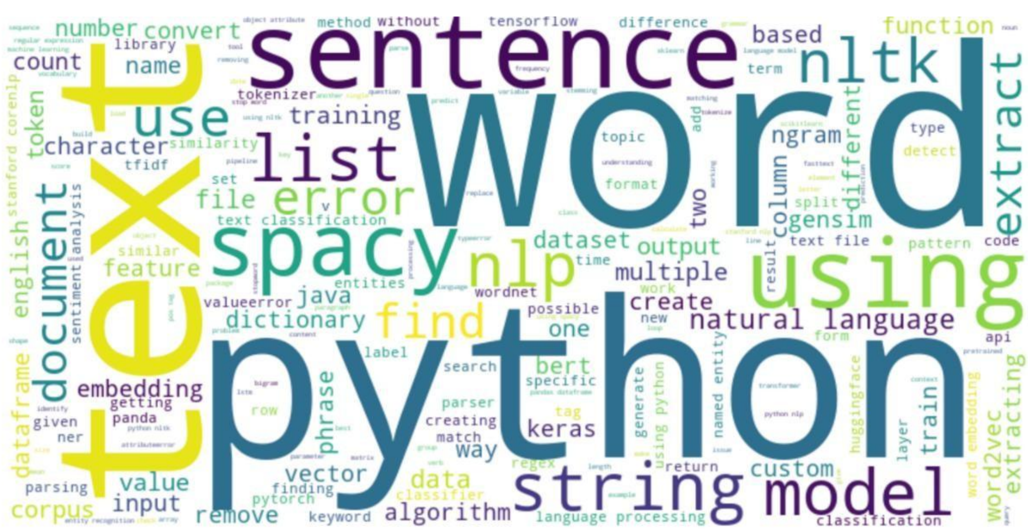
The database organization divided its contents into various subject areas through keyword analysis of post titles. Other category contained the most posts (3250) which showed how various topics were excluded from the predefined groups. The second greatest category of posts involved NLP Library / Tool Usage featuring 1907 postings that showcased discussions about programming with Spacy, NLTK, and Hugging Face. The Data / File Handling category together with Error Debugging joined the list of common areas alongside NLP Library / Tool Usage due to their status as essential programming tasks during troubleshooting. Some of the most popular NLP tasks centered around Information Extraction as well as Similarity / Classification and Text Preprocessing & Cleaning yet fewer community members posted about Model Training / Finetuning as well as Deployment & Optimization and Conceptual / Theory subtopics. Therefore the distribution shows which NLP project help requests come most frequently from users.

	Post ID	Title	Body	Tags	Question Created DateTime	AcceptedAnswerId	Answer Created DateTime	Question Month	Question Day	Question Year	Time_to_Answer_hours
2	79312133	getting leaf words reverse stemming one python...	lines solution provided link trying get leaf w...	python nlp nltk	2024-12-27 15:04:05	79312987.0	2024-12-27 23:39:51	2024-12	2024-12-27	2024	8.596111
7	79298368	inspect probabilities bertopic model	say build bertopic model using bertopic import...	python nlp topic-modeling	2024-12-20 20:49:34	79299703.0	2024-12-21 16:03:22	2024-12	2024-12-20	2024	19.230000

The dataset is after performing all the preprocessing steps and below is the data at its initial stage.

	Post ID	Title	Body	Tags	Question Created DateTime	AcceptedAnswerId	Answer Created DateTime
2	79312133	Getting all leaf words (reverse stemming) into...	<p>On the same lines as the solution provided ...	<python><nlp><nltk>	2024-12-27 15:04:05	79312987.0	2024-12-27 23:39:51
7	79298368	Inspect all probabilities of BERTopic model	Say I build a BERTopic model using \n<pr...	<python><nlp><topic-modeling>	2024-12-20 20:49:34	79299703.0	2024-12-21 16:03:22

Visual data through word clouds depicted the prevalent terms that occurred throughout the Title and Body columns of the database. The size of words inside the word clouds indicates how often they appear in the textual content. Word clouds reveal dominant themes and regular patterns within the text posts by using font size to display frequency occurrences which simplifies the dataset's content distribution for easy examination.



6. Data categorization

The project implementation used rule-based keyword matching to establish post categories from their titles. My system checks all post titles while assigning categories through keyword identification in selected predefined categories. I used my experience in NLP workflows and programming practice combined with knowledge of usual discussions within NLP forums to choose these keywords.

Each classification category corresponded to standard NLP project topics. Posts including error keywords like 'error', 'problem', or 'exception' belonged to the Error Debugging category while NLP Library / Tool Usage labeled titles containing popular libraries 'spacy', 'nltk' and 'bert'. Things categorized as 'file', 'dataframe' and 'json' belonged to the Data / File Handling category and text preprocessing elements such as 'tokenize', 'lemmatize' and 'stopwords' were positioned in Text Preprocessing & Cleaning.

The classification system included information extraction as well as similarity / classification and model training and encompassed conceptual/theoretic and deployment and optimization topics. Material that fell outside other categories received placement under the other section. I selected keywords through a combination of initial dataset analysis and word cloud review and expert's valuable insight. dışmā reached high levels of data organization which revealed the most trending NLP subject areas.

```
Other                             3250
NLP Library / Tool Usage          1907
Data / File Handling              1123
Error Debugging                    646
Information Extraction             613
Similarity / Classification        412
Text Preprocessing & Cleaning      260
Model Training / Fine-tuning       119
Deployment & Optimization          108
Conceptual / Theory                 62
Name: Refined Category, dtype: int64
```

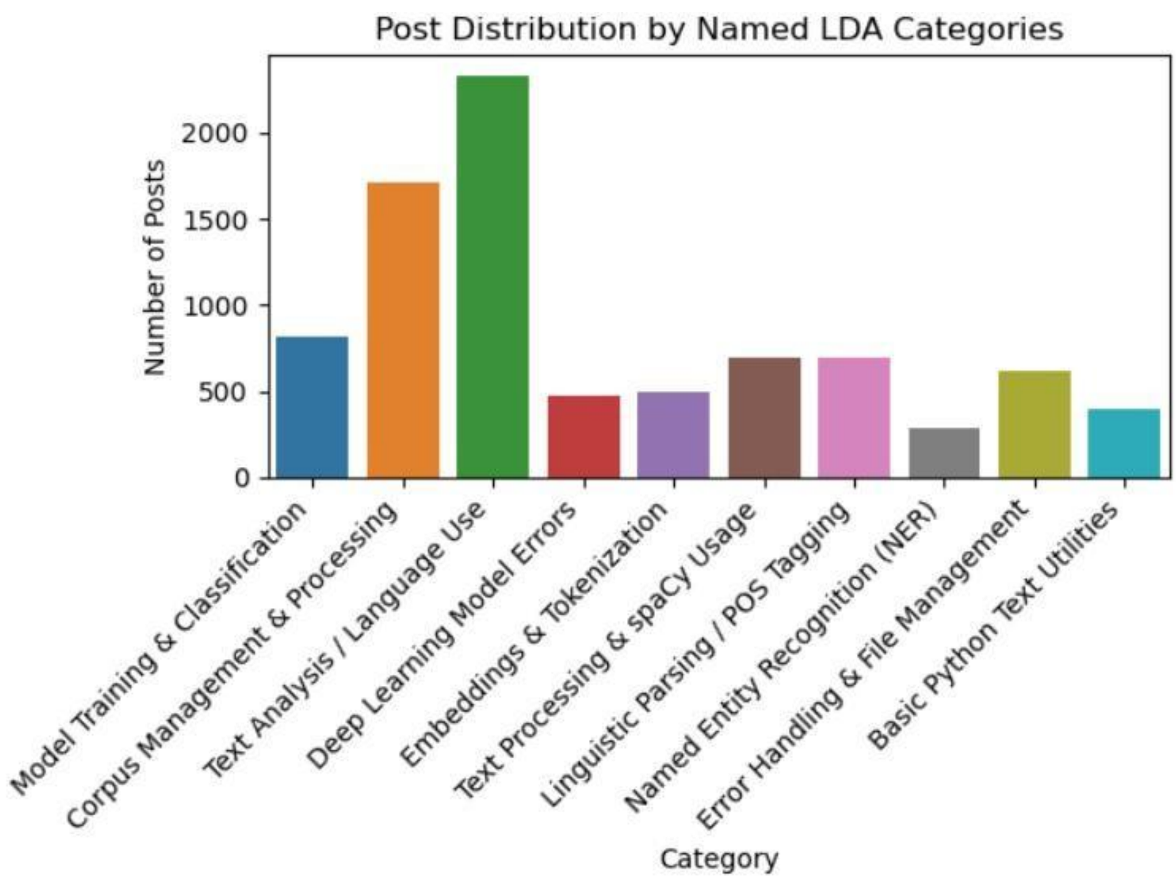
Grouped the dataset through post title keywords into separate topic areas. Three thousand five hundred posts were categorized under the other section since many topics remained outside the predefined groups. Most discussions about NLP library and tool usage accumulated into the 1907 posts of the dataset. The programming tasks of Data / File Handling and Error Debugging appeared frequently which reflects normal programming activities and error debugging needs. The core NLP tasks of Information Extraction and Similarity / Classification and Text

Preprocessing & Cleaning stand out as main categories in the Reddit platform although topics related to Model Training / Fine-tuning, Deployment & Optimization and Conceptual / Theory remain less prevalent. The distribution provides useful information about which parts of NLP projects users require assistance.

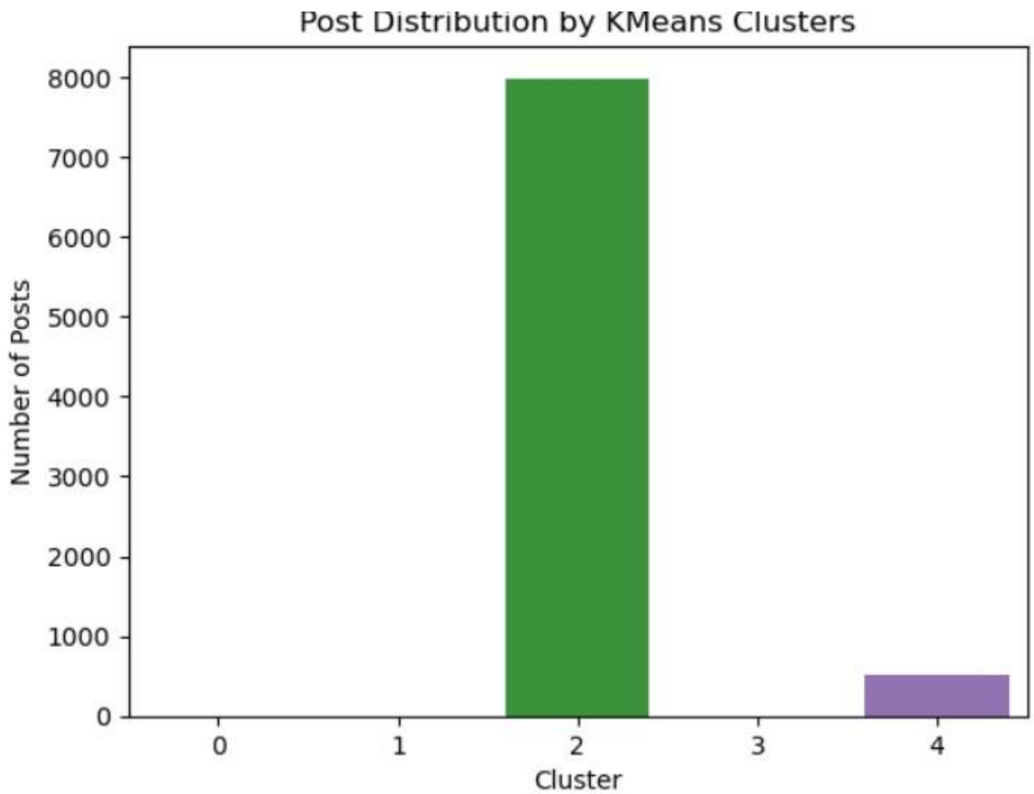
Category	**Example Post**	**SO Link**
Other	determining popular words english dictionary within dictionary words.	link
NLP Library / Tool Usage	Inspect probabilities bertopic model	link
Data / File Handling	getting leaf words reverse stemming one python list	link
Error Debugging	error getting captum text explanations text classification	link
Information Extraction	llama321binstruct generate inconsistent output	link
Similarity / Classification	possible get embeddings nvembed using candle	link
Text Preprocess & cleaning	adjust performance tokenizer	link
Model Training / Fine-tuning	keep training pytorch model new data	link
Deployment & Optimization	capitalized words sentiment analysis	link
Conceptual / Theory	difference multiheadattention attention layer tensorflow	link

An application of Latent Dirichlet Allocation (LDA) identified hidden subject patterns in the textual elements of the data collection. The cleaned text passed to a CountVectorizer that generated a document-term matrix by leaving out English stopwords and removing words that appeared too infrequently or too frequently through min_df and max_df parameter values. The LDA model proceeded to create 10 latent topics through its analysis of word co-occurrence patterns throughout the posts. A unique function presented ten most prominent keywords from each topic so users could understand dominant patterns in their information. The system assigned LDA_Category to each text entry by using the maximum probability scores discovered through the LDA model. A count plot gave visualization of post distribution among analyzed topics.

The research selected LDA because it remains a popular method for topic modeling which reveals latent structures in big unlabeled text collections while remaining interpretable and unsupervised during analysis.



I applied the Elbow Method to find the right number of clusters which would best segment the posts in the dataset through K-Means clustering. I calculated Sum of Squared Errors (SSE) values by repeatedly using cluster numbers from 2 to 20 during the process. The SSE value measures cluster data point relationships which indicates better cluster cohesiveness as values decrease. The sum of squared errors was plotted against cluster numbers by my team during this stage of the analysis. When the decrease in SSE reaches the slowing phase, it is called an elbow point which guides the identification of appropriate cluster numbers that strike an optimal balance between clustering accuracy and model simplicity. This method produced clusters that avoided both simplification of the data through insufficient clusters and prevented difficult interpretation from excessive clustering numbers.



7. Conclusion

My research explored two distinctive text post categorization methods starting with manual title keyword-based classifications followed by an LDA (Latent Dirichlet Allocation) automatic topic discovery. The system allowed me to divide posts into established categories such as Error Debugging along with NLP Library Usage and Data Handling. The unsupervised topic modeling technique LDA served as the basis for my other analysis model. LDA extracted hidden patterns that naturally developed clusters from text content and established three labeled groups: Corpus Management, NER and Deep Learning Errors. The rule-based system produced exact and readable labels, yet LDA showed more extensive finding about fundamental patterns. These methods provided a comprehensive view into the various question and discussion patterns present in the database.

Dataset Link: Link

8. References

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